

Table S10. Person-level evidence synthesis

Person-level evidence synthesis						
	Outcome or analysis scale	Result	FDR-correction set	FDR decision	Exact sample	Interpretation and qualification
Light-exposure behaviour and awareness factors	68 factor-by-metric cells across four participant-level factors	No cell retained FDR support; four sleep-environment cells were non-estimable	68 near-eye factor-by-metric tests	None retained	Metric-specific near-eye samples in Supplementary Table S11	Broad inconclusive result; non-retention is not evidence of no association
Visual light sensitivity	Nine near-eye metrics per one VLSQ-8 SD	Largest displayed ratio shift was melanopic EDI dose, $\times 0.846$ (0.716 to 0.999)	Nine near-eye metric tests	None retained; adjusted $p = 0.160$ for this row	141 participants, 761 participant-days, nine sites for the displayed row	Broad inconclusive result
Corrected midsleep on Free days	Timing outcomes per one-hour later MSFsc	M10 midpoint $+0.205$ h (0.045 to 0.366); L10 midpoint $+0.276$ h (0.118 to 0.435); first light above 250 lx $+0.381$ h (0.146 to 0.616)	Five MCTQ timing outcomes, separate from MEQ	These three rows retained	139 to 140 participants; 722 to 810 participant-days; nine sites	Selective timing associations
Morningness-Eveningness Questionnaire	Timing outcomes per 10 points greater morning preference	M10 midpoint -0.278 h (-0.451 to -0.105); L10 midpoint -0.314 h (-0.487 to -0.142); first light above 250 lx -0.444 h (-0.699 to -0.189)	Five MEQ timing outcomes, separate from MCTQ	These three rows retained	140 to 141 participants; 727 to 816 participant-days; nine sites	Selective timing associations
Age	Near-eye daily metrics per 10 years	Brightest 10 h geometric mean $\times 1.31$ (1.09 to 1.58); time above 1,000 lx $\times 1.27$ (1.11 to 1.45); melanopic EDI dose $\times 1.31$ (1.09 to 1.56)	Near-eye age-metric tests	Three rows retained	141 participants; 761 to 816 participant-days; nine sites	Selective age associations; complementary chest results remain separate
Metric-level biological sex	Near-eye daily metrics, Female minus Male	No near-eye metric-level result retained FDR support	Near-eye biological-sex metric tests	None retained	Metric-specific samples in Supplementary Table S14	Biological sex and gender were recorded separately; gender was not analysed

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	Outcome or analysis scale	Result	FDR-correction set	FDR decision	Exact sample	Interpretation and qualification
Biological-sex-specific daily curve	Complete 24-hour curve and activity-complete sensitivity	The all-available near-eye complete-curve test was supported, but separation attenuated and was not retained in the activity-complete near-eye analysis	Global complete-curve tests and separately prespecified sensitivity decisions	Analysis-specific global decisions	All-available near eye: 141 participants, 816 participant-days, 37,756 observations, nine sites	Attenuation-first interpretation; confidence intervals apply at the displayed clock times

Results that did not meet the applicable FDR criterion are inconclusive rather than evidence of no association. The two chronotype instruments were analysed separately. Biological sex and gender were recorded separately; gender was not analysed.

Supplementary Table S10. Summary of person-level results in the near-eye analyses. Each row identifies the analysis-specific FDR-correction set or decision structure applied to that result. Results that did not meet the applicable FDR criterion are inconclusive rather than evidence of no association. Four light-behaviour sleep-environment cells were non-estimable. The two chronotype instruments were analysed separately. Biological sex and gender were recorded separately; gender was not analysed. The complete available-data curve difference was attenuated and not retained in the activity-complete near-eye analysis.